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June 12, 2026

Editorial Office
Foundations of Physics
Springer Nature

Dear Editors,

I am pleased to submit the manuscript entitled “**Coherence-Dependent Backreaction in Semiclassical and Analog Gravity: Controlled Limits and an Analog-Gravity Test**” for consideration in *Foundations of Physics*.

This paper studies an effective-field-theory extension of semiclassical gravity in which a coarse-grained coherence functional, built from relative entropy, generalized entropy, and local geometric counterterms, generates an informational stress tensor. The framework is presented in controlled near-equilibrium limits rather than as a complete theory of quantum gravity.

The manuscript has two main components. First, it develops the formal construction in settings where the metric variation can be handled explicitly, including a 1+1D conformal toy model and weak-field/Rindler limits. Second, it formulates a concrete analog-gravity test in a Bose-Einstein-condensate sonic horizon. The informational source is fixed by the exact Gaussian-state identity $S(\rho||\sigma_\beta) = \beta_H \Delta E - \Delta S_{\text{vN}}$ (relative entropy as free-energy excess over the Hawking-temperature reference), so both coherent and energy-matched thermal injections source the predicted density modulation at the $\delta\rho/\rho_0 \sim 10^{-6}$ level, and the primary observable is their differential, $\Delta A = \kappa_{\text{eff}} \Delta S_{\text{vN}} \approx 2.3 \times 10^{-7}$ at the baseline, which directly measures the injected von Neumann entropy and is independent of the Hawking temperature at fixed band occupation. The density-response relation is benchmarked to the standard hydrodynamic compressibility limit, to a static Gross-Pitaevskii/Bogoliubov linear-response kernel (0.63% shift of the differential, below 0.9% across one-at-a-time $\pm 25\%$ scans), and to time-dependent 1D Gross-Pitaevskii simulations that validate the kernel at the sub-percent level and quantify the dominant ordinary-nonlinearity systematic. The informational source term itself remains implementation dependent. A differential below 5×10^{-8} in this protocol would falsify the acoustic implementation analyzed in the paper.

I believe the manuscript is appropriate for *Foundations of Physics* because it addresses a foundational question at the interface of semiclassical gravity, quantum information, and open-system dynamics while keeping the central claim empirically vulnerable. The manuscript is careful about scope: it does not claim a UV-complete theory, and it states explicitly that a positive analog result would test the proposed state-geometry feedback structure rather than, by itself, establish direct gravitational backreaction in astrophysical spacetime.

Declarations. This manuscript has not been published elsewhere and is not under consideration by another journal. All work is original. The author has no competing interests to declare. AI tools from Anthropic and OpenAI were used for copy editing and literature synthesis; all scientific content is the author’s own.

Thank you for considering this submission.

Sincerely,

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